PUBLIC PROCUREMENT DOCUMENT COMPLETENESS AGENT

Workflow: Sense/Context → Plan/Decide → Act/Tool → Observe → Stop/Re-plan

1. Purpose

This workflow defines how the Public Procurement Document Completeness Agent processes an uploaded procurement document in a controlled agentic loop. The agent understands the task, chooses an action, uses an approved tool, observes the result, and either stops or re-plans. The workflow is bounded so that the agent cannot continue indefinitely or act outside document-completeness checking.

2. Sense / Context

The agent receives the procurement document and establishes the context required for the task. It checks that a document has been provided, determines whether it can be processed, loads the applicable completeness checklist, and initializes its working state. The state can contain the uploaded document, extracted content, checklist, findings, errors and attempt count.

Decision: If no valid document is available, stop and request a valid document.

3. Plan / Decide

The agent examines its current state and selects the next necessary action. The normal plan is: extract document content → compare the content with the completeness checklist → collect findings → produce the completeness result. The agent selects only an approved action that is necessary for this goal.

4. Act / Tool

The agent executes the selected action using an approved tool. For example, the first tool extracts usable content from the procurement document. The second tool checks that content against the required procurement completeness checklist.

```
extract_document_content(document)
```

```
check_document_completeness(document_content, completeness_checklist)
```

5. Observe

After each tool call, the agent examines the returned result. It checks whether the action succeeded, whether usable content was extracted, which requirements were found or not found, whether any result is uncertain, and whether an error occurred. The tool result is recorded in the current state before the agent makes another decision.

6. Stop / Re-plan

If all required checks have been completed with sufficient evidence, the agent stops calling tools and produces the completeness report. If an action fails or the evidence is insufficient, the agent may re-plan and perform another permitted action. Re-planning is allowed only while the workflow remains within its defined bounds.

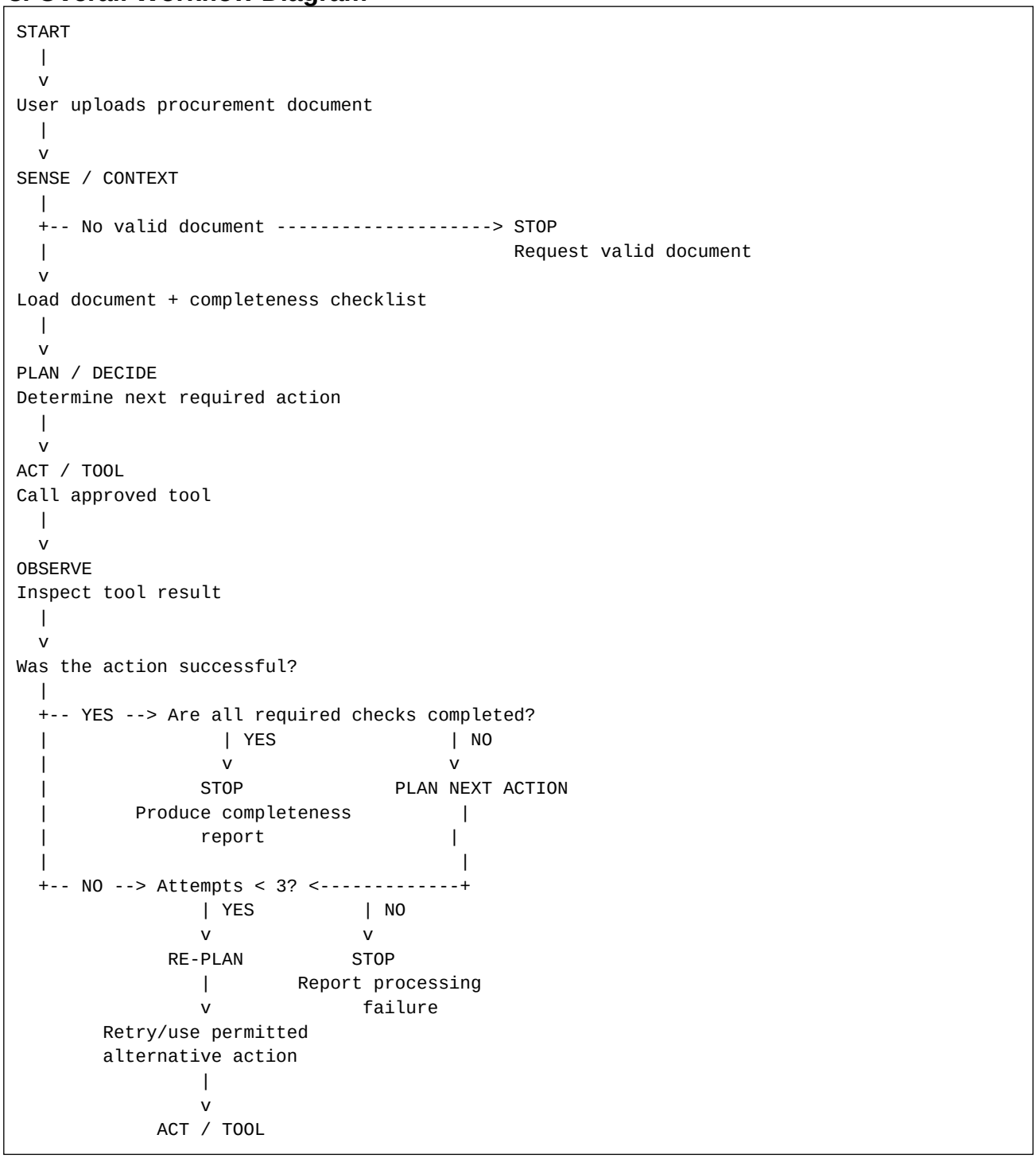
Stop example: The agent successfully checks eight requirements and finds six present and two not found. It reports those findings and stops.

Re-plan example: Document extraction fails. The agent observes the failure, checks the attempt count, and if the maximum has not been reached, retries or uses another permitted processing approach. If the limit is reached, it stops and reports that reliable processing could not be completed.

7. Workflow Bounds

Boundary	Rule
Maximum attempts	Maximum of 3 action/re-plan attempts for processing failures.
Tool restriction	Only approved tools for procurement-document completeness checking may be used.
Scope restriction	The agent checks document completeness only; it does not make procurement award or approval decisions.
No fabrication	The agent must never invent information that is absent from the document.
Uncertainty	Information that cannot be reliably verified is reported as uncertain or not verified.
Failure handling	When the maximum attempt limit is reached, the agent stops and reports the failure.
Completion	The agent stops once all required completeness checks have been performed.
Human control	Final procurement decisions remain with the responsible human/procurement officer.

8. Overall Workflow Diagram



9. Pseudocode

```
MAX_ATTEMPTS = 3

run_agent(document, checklist):
    attempts = 0

    SENSE:
        if document is invalid:
            STOP and request a valid document

        initialize state

    while attempts < MAX_ATTEMPTS:

        PLAN:
            if content is missing:
                action = extract_content
            elif findings are missing:
                action = check_completeness
            else:
                STOP and return completeness report

        ACT:
            attempts = attempts + 1
            execute selected approved tool

        OBSERVE:
            inspect tool result
            update state

        if action failed:
            RE-PLAN and continue

    STOP:
        report that maximum permitted attempts were reached
```

10. Expected Output

When successful, the workflow produces a completeness result showing the requirements checked, requirements found, requirements not found, and any items that could not be reliably verified. When unsuccessful, it stops safely and reports why processing could not be completed.

11. Conclusion

This is a bounded agent workflow because the agent can make decisions and re-plan based on observations, but it operates within explicit limits. It has a restricted purpose, approved tools, a maximum number of attempts, clear failure behaviour, and defined stop conditions. This prevents uncontrolled looping and unsupported procurement conclusions.